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TG-DCM-PINN: Target-Gated Direction-Cosine Physics for Six-Axis Inertial Attitude Estimation

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PROBLEM

Pointwise supervision can produce accurate individual tilt estimates without enforcing gyro-consistent motion between frames, while accelerometer-based correction becomes unreliable when dynamic acceleration contaminates gravity.

RESEARCH QUESTION

Can a learned estimator improve tilt by enforcing inertial relationships only when the measurements support them?

CONTRIBUTIONS

1 Direction-cosine tilt physics

Convert each predicted quaternion into its body-frame vertical direction, so every physical term acts directly on the observable tilt geometry.

2 Target-gated gyro dynamics

Use labelled transitions to identify when gyro propagation is trustworthy, then enforce prediction consistency only on accepted pairs.

3 Confidence-gated gravity

Build a continuous confidence from acceleration magnitude, jerk, and angular rate before applying the gravity anchor.

4 Robust composite objective

Combine supervised tilt, unit-quaternion regularization, two gated physics terms, dead zones, and Charbonnier penalties.

5 Regime-aware evidence

Evaluate 559 sequences and expose dataset, motion-family, confidence-interval, and tail behavior instead of reporting one pooled mean.

WHY CHARBONNIER?

$$\rho_\varepsilon(r) = \sqrt{r^2 + \varepsilon^2} - \varepsilon$$

Small residuals receive a smooth quadratic penalty; large residuals grow linearly. Difficult or corrupted samples therefore cannot dominate the

Near zero $\rho_\varepsilon(r) \approx \frac{r^2}{2\varepsilon}$

Large residual $\rho_\varepsilon(r) \approx |r|$

PHYSICS-INFORMED REGULARIZER

The DCM appears only inside the auxiliary loss. Each normalized quaternion is mapped to the body-frame world-vertical direction h(q). Dynamics compares the gyro-propagated prediction with the next prediction; gravity compares h(q) with normalized acceleration. Both terms constrain the observable tilt direction without constructing a full rotation matrix.

METHOD

IMU window
acceleration · angular rate · Δt

→

Neural estimator
2-layer GRU · hidden 200

→

Prediction
quaternion → vertical direction

Vertical direction used by both gates

$$h(q) = [2(xz - wy),\ 2(yz + wx),\ 1 - 2(x^2 + y^2)]^\top$$

A TARGET-GATED GYRO DYNAMICS

1. Propagate the labelled tilt with the measured angular increment.

$$q^g_{t+1} = q_t \otimes \mathrm{Exp}(\omega_t \Delta t)$$

2. Accept the transition only when propagation matches the next labelled tilt.

$$m_{\mathrm{dyn}} = \mathbf{1}\left[e_{\mathrm{tilt}}(q^g_{t+1}, q_{t+1}) \le 2^\circ\right]$$

3. For accepted pairs, propagate the prediction and penalize disagreement between propagated and next predicted vertical directions. Ignore the first 0.15°.

Purpose: enforce measured rotational evolution only on trustworthy target transitions.

B CONFIDENCE-GATED GRAVITY

1. Estimate whether acceleration is a credible gravity observation.

$$m_{\mathrm{grav}} = e^{-\frac{(\|a_t\| - g)^2}{2\sigma_a^2}} e^{-\frac{j_t^2}{2\sigma_j^2}} e^{-\frac{\|\omega_t\|^2}{2\sigma_\omega^2}}$$

Acceleration magnitude checks proximity to gravity; jerk detects changing linear acceleration; angular rate suppresses correction during fast rotation.

2. Align predicted vertical with normalized acceleration using this continuous weight.

$$L_{\mathrm{grav}} = \mathrm{mean}\left[m_{\mathrm{grav}}\, \rho\left(\angle\left(h(\hat{q}_t), \tfrac{a_t}{\|a_t\|}\right) - 0.20^\circ\right)_+\right]$$

Purpose: use gravity strongly in calm motion and suppress it during dynamic acceleration.

COMPLETE TRAINING OBJECTIVE

$$L = L_{\mathrm{sup}} + 0.01 L_{\mathrm{norm}} + 0.10 L_{\mathrm{dyn}} + 0.05 L_{\mathrm{grav}}$$

Dead zones: dynamics 0.15° · gravity 0.20° · Charbonnier scale 0.50°

IMPLEMENTED GATED TRAINING LOGIC

SEQUENCE INPUT
acceleration · angular rate · Δt
training target quaternion

→ **RECURRENT ESTIMATOR**
2-layer GRU · hidden 200
predicted quaternion sequence

→ **DCM VERTICAL ROW**
normalize q, then compute h(q)
shared tilt representation

→ **COMPOSITE OBJECTIVE**
supervised tilt + norm
conditional dynamics + gravity

ALWAYS ACTIVE ON ITS VALID SUPPORT Labelled frames: supervised tilt loss. Every prediction: unit-quaternion regularizer weighted 0.01.

A DYNAMICS BRANCH · PAIR-LEVEL · TARGET-GATED

PAIR VALID?
two finite labels
finite gyro and Δt

→ **TARGET TEST**
gyro-propagate target
accept only if error ≤ 2°

→ **PREDICTION TEST**
gyro-propagate prediction
compare vertical directions

→ **ROBUST RESIDUAL**
remove first 0.15°
Charbonnier scale 0.50°

→ **BRANCH OUTPUT**
masked mean × 0.10
failed decision → exact zero

B GRAVITY BRANCH · FRAME-LEVEL · CONFIDENCE-GATED

INPUT VALID?
finite acceleration, gyro and Δt

→ **THREE FACTORS**
|a| near 9.81 · low jerk
low angular rate

→ **GATE DECISION**
multiply confidences
accept only if > 10^{-4}

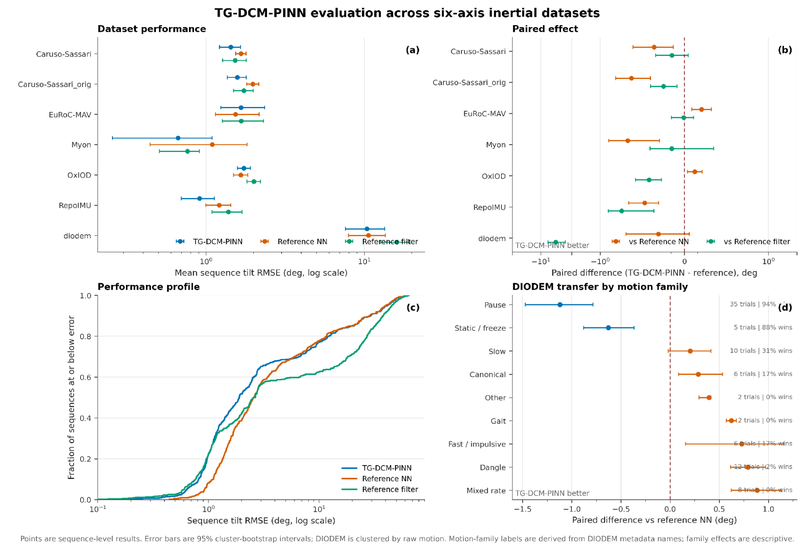
→ **TILT RESIDUAL**
h(q) versus a/|a|
remove first 0.20°

→ **BRANCH OUTPUT**
confidence × Charbonnier
masked mean × 0.05

LOSS EXECUTION Active terms are summed and backpropagated; an empty mask returns zero instead of an invalid mean. The target gate is binary. The gravity gate is binary at the floor and continuous above it.

EVALUATION

Primary metric: sequence-level tilt RMSE. Paired bootstrap intervals use sequences on the standard split and raw-motion clusters on DIODEM.



Points are sequence-level results. Error bars are 95% cluster-bootstrap intervals; DIODEM is clustered by raw motion. Motion-family labels are derived from DIODEM metadata names; family effects are descriptive.

Dataset means, paired effects, performance profile, and DIODEM motion families. Negative paired change favors TG-DCM.

STANDARD SPLIT

Mean tilt RMSE decreases by 0.091°. The paired interval excludes zero.

DATASET DEPENDENCE

Caruso-Sassari, Myon, and RepoIMU improve; EuRoC-MAV and OxIOD regress.

DIODEM TRANSFER

Mean change is −0.195°, but the raw-motion-clustered interval crosses zero.

MOTION DEPENDENCE

Pause and static segments improve most. Sustained and impulsive motion remains the weak regime.

CONCLUSION

Reliability-gated physical constraints improve average standard-domain tilt accuracy. The gain concentrates in low-dynamics regimes; the next method must learn when and how strongly to correct tilt during sustained and abrupt motion.

SCOPE AND RESEARCH BOUNDARY

TRAINING-TIME PHYSICS, LEAN DEPLOYMENT

The DCM projection, target validation, confidence calculation, and both physics losses exist only in training. Deployment receives acceleration, angular rate, Δt, and recurrent state; it returns a normalized quaternion. No target, gate, or separate physical filter is required.

LIMITATIONS SUPPORTED BY CODE AND RESULTS

The three confidence scales and 10^{-4} floor are fixed globally, not calibrated to sensor or motion regime. The target gate needs consecutive valid labels. Low-jerk sustained acceleration near |a| = g can remain plausible to the gravity gate. No controlled ablation isolates either physics term or the training stack. DIODEM remains heavy-tailed: median 2.70°, p95 44.20°, maximum 62.38°.

NEXT EXPERIMENTS

Run controlled multi-seed ablations for supervised-only, dynamics-only, gravity-only, dead zones, confidence factors, training budget, and EMA. Replace fixed gravity confidence with a temporally calibrated estimator that models sensor bias and sustained specific force while retaining physically interpretable inputs. Select and report models with motion-balanced and tail-risk criteria, not mean tilt alone.

TRAINING

- 512 epochs, 300 batches/epoch, 9000-sample windows, TBPTT length 1800.
- Ranger; gradient clipping 1.0; BF16/TF32 on CUDA
- EMA checkpoint selected at epoch 491.
- Val: 557 sequences

INTERPRETATION

- The dynamics gate is label-derived: it tests whether gyro propagation agrees with the next target.
- The gravity gate is input-derived: it estimates confidence from acceleration magnitude, jerk, and angular rate.
- Gate values regulate training gradients; they are not evaluation-time quality scores.

EXACT ZERO-CONTRIBUTION CASES

- Dynamics is zero for fewer than two frames, invalid consecutive targets, non-finite gyro/timing, or target-propagation error above 2°.
- Gravity is zero for non-finite inputs or combined confidence at or below 10^{-4}; accepted frames retain continuous confidence weighting.
- Residuals inside the 0.15° dynamics or 0.20° gravity dead zone are zero before robustification.

TG-DCM-PINN treats physics as conditional evidence: validates gyro transitions | estimates gravity reliability | applies each tilt constraint only where it is credible.

Observed result: lower mean tilt error on the standard split, with remaining failures concentrated in dynamic motion.